

Just as before, the non-covariant effects of the second term may be removed by adding to the interaction a non-covariant term

$$\mathcal{H}_{NC}(x) = \frac{1}{2} [J^0(x)]^2, \quad (6.2.27)$$

where $J^\mu(x)$ is here the current which multiplies $\partial_\mu \phi(x)$ in the covariant part of $\mathcal{H}(x)$.

It should be clear that (at least for massive particles) the effects of non-covariant parts of the propagator can always be cancelled in this way by adding non-covariant local terms to the Hamiltonian density. This is because the numerator $P_{\ell m}^{(L)}(q)$ in the propagator must equal the covariant polynomial $P_{\ell m}(q)$ when q^μ is on the mass shell, so the difference between $P_{\ell m}^{(L)}(q)$ and $P_{\ell m}(q)$ must contain a factor $q^2 + m^2$. This factor cancels the denominator $(q^2 + m^2 - i\epsilon)$ in the contribution of this difference to Eq. (6.2.18), so Eq. (6.2.18) always equals a covariant term plus a term proportional to the delta function $\delta^4(x - y)$ or its derivatives. The effect of the latter term may be cancelled by adding to the interaction a term quadratic in the currents to which the paired fields couple, or in their derivatives. In what follows, it will be assumed tacitly that such a term has been included in the interaction, and in consequence we shall use the *covariant* polynomial $P_{\ell m}(q)$ in the propagator (6.2.18), and will thus henceforth drop the label 'L'.

It may seem that this is a rather *ad hoc* procedure. Fortunately, in the canonical formalism discussed in the following chapter, the non-covariant term in the Hamiltonian density needed to cancel non-covariant terms in the propagator arises automatically. This, in fact, forms part of the motivation for introducing the canonical formalism.

* * *

Before closing this section, it may be useful to mention some other definitions of the propagator, equivalent to Eq. (6.2.1), that appear commonly in the literature. First, taking the vacuum expectation value of Eq. (6.1.14) gives

$$\begin{aligned} -i\Delta_{\ell m}(x, y) = & \theta(x - y) \left\langle \left[\psi_\ell^+(x), \psi_m^{+\dagger}(y) \right]_{\mp} \right\rangle_0 \\ & \pm \theta(y - x) \left\langle \left[\psi_m^{-\dagger}(y), \psi_\ell^-(x) \right]_{\mp} \right\rangle_0. \end{aligned} \quad (6.2.28)$$

(Here $\langle AB \cdots \rangle_0$ denotes the vacuum expectation value $(\Phi_0, AB \cdots \Phi_0)$.) Both $\psi_\ell^+(x)$ and $\psi_m^{-\dagger}(y)$ annihilate the vacuum, so only one term in each commutator or anticommutator in Eq. (6.2.28) actually contributes to the propagator:

$$-i\Delta_{\ell m}(x, y) = \theta(x - y) \langle \psi_\ell^+(x) \psi_m^{+\dagger}(y) \rangle_0 \pm \theta(y - x) \langle \psi_m^{-\dagger}(y) \psi_\ell^-(x) \rangle_0. \quad (6.2.29)$$